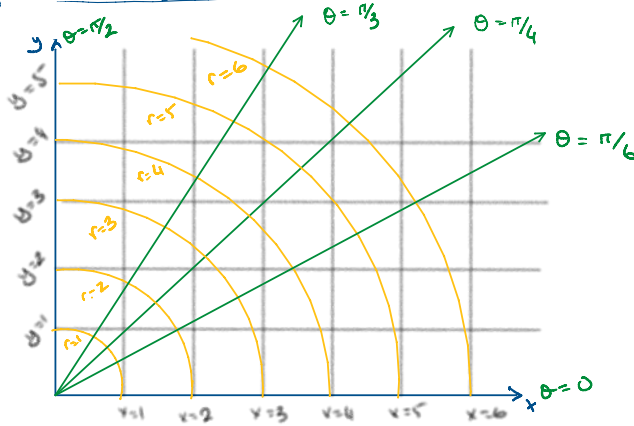


22 Case Study: Cartesian and Polar Coordinates

22.1



We know:

Transformation/Transition functions:

$$\begin{aligned} x &= r \cos \theta \\ y &= r \sin \theta \end{aligned} \quad \left| \quad \begin{aligned} r &= \sqrt{x^2 + y^2} \\ \theta &= \arctan\left(\frac{y}{x}\right) \end{aligned} \right.$$

22.2

Focus on:

$$x = r \cos \theta$$

$$y = r \sin \theta$$

We can use the partials

$$\frac{\partial x}{\partial r} = \cos \theta \quad \frac{\partial x}{\partial \theta} = -r \sin \theta$$

$$\frac{\partial y}{\partial r} = \sin \theta \quad \frac{\partial y}{\partial \theta} = r \cos \theta$$

To create the Derivative matrix:

$$D = \begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{bmatrix} = \begin{bmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{bmatrix}$$

The determinant of this derivative matrix is called the Jacobian

$$J(r, \theta) = \det(D) = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r \cos^2 \theta - (-r \sin^2 \theta) \\ = r (\cos^2 \theta + \sin^2 \theta) = r$$

Slide: What Jacobian do we get if we use $D = \begin{bmatrix} \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial r} \\ \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial r} \end{bmatrix}$?

What is the significance?

Notation: We use $\frac{\partial(x, y)}{\partial(r, \theta)}$ to indicate the absolute value of the Jacobian

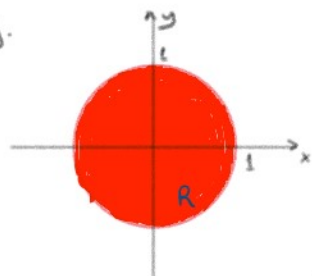
$$\text{So here: } \frac{\partial(x, y)}{\partial(r, \theta)} = |r| = r.$$

22.3 Change of variables: Double Integrals

Suppose $f(x, y)$ is a scalar field. Then

$$\iint_{R(x, y)} f(x, y) dx dy = \iint_{R(r, \theta)} \underbrace{f(x(r, \theta), y(r, \theta))}_{\text{write it in terms of } r, \theta} \frac{\partial(x, y)}{\partial(r, \theta)} dr d\theta$$

eg.



$$\begin{aligned} \iint_R (-2x - 2y - 1) dx dy &= \iint_R [-2r \cos \theta - 2r \sin \theta - 1] \underbrace{r}_{\frac{\partial(x, y)}{\partial(r, \theta)}} dr d\theta \\ &= \int_{\theta=0}^{2\pi} \int_{r=0}^1 [-2r(\cos \theta + \sin \theta) - 1] r dr d\theta = -\pi. \end{aligned}$$

22.4 Change of variables: line element

We also often define an object called the line element:

$$ds^2 = dx^2 + dy^2$$

Then in (r, θ) -coordinates:

$$ds^2 = \left[\left(\frac{\partial x}{\partial r} \right)^2 + \left(\frac{\partial y}{\partial r} \right)^2 \right] dr^2 + \left[\left(\frac{\partial x}{\partial \theta} \right)^2 + \left(\frac{\partial y}{\partial \theta} \right)^2 \right] d\theta^2$$

$$ds^2 = \left[(\cos^2 \theta) + (\sin^2 \theta) \right] dr^2 + \left[(-r \sin \theta)^2 + (r \cos \theta)^2 \right] d\theta^2$$

$$\Rightarrow ds^2 = \underbrace{(1)}_{h_1^2} dr^2 + \underbrace{(r^2)}_{h_2^2} d\theta^2$$

22.5 Lamé Coefficients:

$$h_1 = \left| \begin{bmatrix} \frac{\partial x}{\partial r} \\ \frac{\partial y}{\partial r} \end{bmatrix} \right| = \sqrt{\left(\frac{\partial x}{\partial r} \right)^2 + \left(\frac{\partial y}{\partial r} \right)^2} = \sqrt{1} = 1$$

$$h_2 = \left| \begin{bmatrix} \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial \theta} \end{bmatrix} \right| = \sqrt{\left(\frac{\partial x}{\partial \theta} \right)^2 + \left(\frac{\partial y}{\partial \theta} \right)^2} = \sqrt{r^2} = r$$

These play a very important role in the remaining discussions

22.6 Change of variables: gradient

$$\text{Recall: } \vec{\nabla} = \left[\frac{\partial}{\partial x} \right] = \frac{\partial}{\partial r} \hat{r} + \frac{\partial}{\partial \theta} \hat{\theta}$$

22.6 Change of variables: Gradient

Recall: $\vec{\nabla} = \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{bmatrix} = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j}$

So that: $\vec{\nabla} f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j}$

Now, recall that the position vector:

$$\vec{r} = x \hat{i} + y \hat{j}$$

This can also be written as

$$\vec{r} = r \hat{e}_r$$

An interesting connection:

$$\hat{i} = \frac{\partial \vec{r}}{\partial x} \quad \hat{j} = \frac{\partial \vec{r}}{\partial y}$$

Similarly: $\hat{e}_r = \frac{\partial \vec{r}}{\partial r}$

Now, using Chain Rule:

$$\hat{e}_r = \frac{\partial \vec{r}}{\partial r} = \frac{\partial}{\partial r} (x \hat{i} + y \hat{j}) = \left(\frac{\partial x}{\partial r} \right) \hat{i} + \left(\frac{\partial y}{\partial r} \right) \hat{j}$$

$$\Rightarrow \hat{e}_r = \cos \theta \hat{i} + \sin \theta \hat{j}$$

What about $\hat{e}_\theta$?

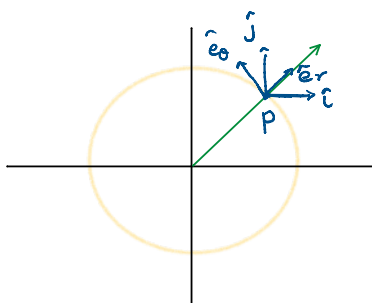
In general, $\hat{e}_\theta$ is tangent to $r = \text{const.}$

From the parametrization $x = r \cos \theta$, $y = r \sin \theta$ [$r = \text{const.}$], we get the tangent vector

$$\begin{bmatrix} dx/d\theta \\ dy/d\theta \end{bmatrix} = \begin{bmatrix} -r \sin \theta \\ r \cos \theta \end{bmatrix} \leftarrow \text{Slide: What is the length?}$$

Now $\hat{e}_\theta$ is the unit vector in this direction, so:

$$\hat{e}_\theta = -\sin \theta \hat{i} + \cos \theta \hat{j}$$



So then:

$$\left. \begin{aligned} \hat{e}_r &= \cos \theta \hat{i} + \sin \theta \hat{j} \\ \hat{e}_\theta &= -\sin \theta \hat{i} + \cos \theta \hat{j} \end{aligned} \right\} \Rightarrow \begin{bmatrix} \hat{e}_r \\ \hat{e}_\theta \end{bmatrix} = \underbrace{\begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}}_{\text{What is this?}} \begin{bmatrix} \hat{i} \\ \hat{j} \end{bmatrix}$$

Which allows us to say:

$$\begin{bmatrix} \hat{i} \\ \hat{j} \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}^{-1} \begin{bmatrix} \hat{e}_r \\ \hat{e}_\theta \end{bmatrix}$$

$$\begin{bmatrix} \hat{i} \\ \hat{j} \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}^{-1} \begin{bmatrix} \hat{e}_r \\ \hat{e}_\theta \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \hat{i} \\ \hat{j} \end{bmatrix} = \frac{1}{\cos^2\theta + \sin^2\theta} \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \hat{e}_r \\ \hat{e}_\theta \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \hat{i} \\ \hat{j} \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \hat{e}_r \\ \hat{e}_\theta \end{bmatrix}$$

This can be summarized as:

$$\hat{i} = \cos\theta \hat{e}_r - \sin\theta \hat{e}_\theta$$

$$\hat{j} = \sin\theta \hat{e}_r + \cos\theta \hat{e}_\theta$$

Back to gradient.

$$\vec{\nabla} f = \left(\frac{\partial f}{\partial x} \right) \hat{i} + \left(\frac{\partial f}{\partial y} \right) \hat{j}$$

$$= \left[\frac{\partial f}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial f}{\partial \theta} \frac{\partial \theta}{\partial x} \right] [\cos\theta \hat{e}_r - \sin\theta \hat{e}_\theta] + \left[\frac{\partial f}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial f}{\partial \theta} \frac{\partial \theta}{\partial y} \right] [\sin\theta \hat{e}_r + \cos\theta \hat{e}_\theta]$$

Note: From $r = \sqrt{x^2 + y^2}$, we get

$$\frac{\partial r}{\partial x} = \frac{\frac{\partial}{\partial x} \sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2}} = \frac{x}{r} = \frac{r \cos\theta}{r} = \cos\theta$$

$$\text{and } \frac{\partial r}{\partial y} = \frac{y}{r} = \sin\theta$$

Similarly: $\theta = \tan^{-1}(y/x)$, so

$$\frac{\partial \theta}{\partial x} = \frac{1}{1 + (y/x)^2} \left(-\frac{y}{x^2} \right) = \frac{-y}{x^2 + y^2} = \frac{-y}{r^2} = \frac{-r \sin\theta}{r^2} = -\frac{1}{r} \sin\theta$$

$$\frac{\partial \theta}{\partial y} = \frac{1}{1 + (y/x)^2} \left(\frac{1}{x} \right) = \frac{x}{x^2 + y^2} = \frac{r \cos\theta}{r^2} = \frac{1}{r} \cos\theta$$

So:

$$\vec{\nabla} f = \left[\frac{\partial f}{\partial r} (\cos\theta) + \frac{\partial f}{\partial \theta} \left(-\frac{1}{r} \sin\theta \right) \right] [\cos\theta \hat{e}_r - \sin\theta \hat{e}_\theta]$$

$$+ \left[\frac{\partial f}{\partial r} (\sin\theta) + \frac{\partial f}{\partial \theta} \left(\frac{1}{r} \cos\theta \right) \right] [\sin\theta \hat{e}_r + \cos\theta \hat{e}_\theta]$$

$$= \hat{e}_r \left[\frac{\partial f}{\partial r} (\cos\theta) \cos\theta + \frac{\partial f}{\partial r} (\sin\theta) \sin\theta + \frac{\partial f}{\partial \theta} \left(-\frac{1}{r} \sin\theta \right) (-\sin\theta) + \frac{\partial f}{\partial \theta} \left(\frac{1}{r} \cos\theta \right) \cos\theta \right]$$

$$+ \hat{e}_\theta \left[\frac{\partial f}{\partial r} (\cos\theta) (-\sin\theta) + \frac{\partial f}{\partial r} (\sin\theta) (\cos\theta) + \frac{\partial f}{\partial \theta} \left(-\frac{1}{r} \sin\theta \right) \cos\theta + \frac{\partial f}{\partial \theta} \left(\frac{1}{r} \cos\theta \right) (-\sin\theta) \right]$$

$$= \hat{e}_r \left[\frac{\partial f}{\partial r} \cos\theta + \frac{\partial f}{\partial r} \sin\theta \right] + \hat{e}_\theta \left[\frac{1}{r} \frac{\partial f}{\partial \theta} \sin\theta + \frac{1}{r} \frac{\partial f}{\partial \theta} \cos\theta \right]$$

$$= \hat{e}_r \left[\frac{\partial}{\partial r} \cos\theta + \frac{\partial}{\partial r} \sin\theta \right] + \hat{e}_\theta \left[r \frac{\partial}{\partial \theta} + \frac{1}{r} \frac{\partial}{\partial \theta} \right]$$

$$= \left(\frac{\partial f}{\partial r} \right) \hat{e}_r + \frac{1}{r} \left(\frac{\partial f}{\partial \theta} \right) \hat{e}_\theta$$

In other words:

$$\vec{\nabla} = \hat{e}_r \frac{\partial}{\partial r} + \hat{e}_\theta \frac{1}{r} \frac{\partial}{\partial \theta} \quad \text{in polar coordinates}$$

22.7 Using Lamé coefficients and change of basis:

Note that:

$$\hat{e}_r = \cos\theta \hat{i} + \sin\theta \hat{j} = \underbrace{\left(\frac{\partial x}{\partial r} \right)}_{h_1} \hat{i} + \underbrace{\left(\frac{\partial y}{\partial r} \right)}_{h_2} \hat{j}$$

Lamé coeff

$$\hat{e}_\theta = -\sin\theta \hat{i} + \cos\theta \hat{j} = \frac{1}{r} \frac{\partial x}{\partial \theta} \hat{i} + \frac{1}{r} \frac{\partial y}{\partial \theta} \hat{j}$$

Similarly:

$$\hat{i} = \cos\theta \hat{e}_r - \sin\theta \hat{e}_\theta = \underbrace{\left(\frac{\partial r}{\partial x} \right)}_{h_1} \hat{e}_r + \underbrace{\left(r \frac{\partial \theta}{\partial x} \right)}_{h_2} \hat{e}_\theta$$

$$\hat{j} = \sin\theta \hat{e}_r + \cos\theta \hat{e}_\theta = \left(\frac{\partial r}{\partial y} \right) \hat{e}_r + \left(r \frac{\partial \theta}{\partial y} \right) \hat{e}_\theta$$

Lamé coeff

and in polar coordinates:

$$\vec{\nabla} = \left(\frac{\partial}{\partial r} \right) \hat{e}_r + \left(\frac{1}{r} \frac{\partial}{\partial \theta} \right) \hat{e}_\theta$$

Lamé coefficients

22.8 In general, for any vector field

$$\vec{F} = F_1(x, y) \hat{i} + F_2(x, y) \hat{j}$$

in polar coordinates:

$$\vec{F} = F_1(x(r, \theta), y(r, \theta)) \left[h_1 \frac{\partial r}{\partial x} \hat{e}_r + h_2 \frac{\partial \theta}{\partial x} \hat{e}_\theta \right]$$

$$+ F_2(x(r, \theta), y(r, \theta)) \left[h_1 \frac{\partial r}{\partial y} \hat{e}_r + h_2 \frac{\partial \theta}{\partial y} \hat{e}_\theta \right]$$

$$= \left[F_1 h_1 \frac{\partial r}{\partial x} + F_2 h_1 \frac{\partial r}{\partial y} \right] \hat{e}_r + \left[F_1 h_2 \frac{\partial \theta}{\partial x} + F_2 h_2 \frac{\partial \theta}{\partial y} \right] \hat{e}_\theta$$

$$= \underbrace{\left[F_1 h_1 \frac{\partial r}{\partial x} + F_2 h_1 \frac{\partial r}{\partial y} \right]}_{F_r} \hat{e}_r + \underbrace{\left[F_1 h_2 \frac{\partial \theta}{\partial x} + F_2 h_2 \frac{\partial \theta}{\partial y} \right]}_{F_\theta} \hat{e}_\theta$$

$$\text{So: } F_r = F_1 h_1 \frac{\partial r}{\partial x} + F_2 h_1 \frac{\partial r}{\partial y} = F_1(x(r, \theta), y(r, \theta)) (1) (\cos \theta) + F_2(x(r, \theta), y(r, \theta)) (1) (\sin \theta)$$

$$\Rightarrow F_r = F_1 \cos \theta + F_2 \sin \theta$$

$$\begin{aligned} \text{and } F_\theta &= F_1 h_2 \frac{\partial \theta}{\partial x} + F_2 h_2 \frac{\partial \theta}{\partial y} \\ &= F_1(r) \left(-\frac{1}{r} \sin \theta \right) + F_2(r) \left(\frac{1}{r} \cos \theta \right) \\ &= -F_1 \sin \theta + F_2 \cos \theta \end{aligned}$$

$$\text{i.e. } \vec{F} = [F_1 \cos \theta + F_2 \sin \theta] \hat{e}_r + [-F_1 \sin \theta + F_2 \cos \theta] \hat{e}_\theta$$

$$\text{e.g. let } \vec{F} = \begin{bmatrix} x^2 + y^2 \\ -2y \end{bmatrix} = (x^2 + y^2) \hat{i} - 2y \hat{j}$$

$$\begin{aligned} \text{Then: } F_1 &= x^2 + y^2 = r^2 \\ F_2 &= -2y = -2r \sin \theta \end{aligned}$$

$$\text{and } F_r = F_1 \cos \theta + F_2 \sin \theta = r^2 \cos \theta - 2r \sin \theta$$

$$F_\theta = -F_1 \sin \theta + F_2 \cos \theta = -r^2 \sin \theta - 2r \sin \theta \cos \theta$$

22.9 Divergence in Polar Coordinates:

$$\vec{\nabla} \cdot \vec{F} = \frac{\partial}{\partial x} F_1 + \frac{\partial}{\partial y} F_2$$

$$= \left[\frac{\partial r}{\partial x} \frac{\partial}{\partial r} + \frac{\partial \theta}{\partial x} \frac{\partial}{\partial \theta} \right] F_1 + \left[\frac{\partial r}{\partial y} \frac{\partial}{\partial r} + \frac{\partial \theta}{\partial y} \frac{\partial}{\partial \theta} \right] F_2$$

$$= \left[\cos \theta \frac{\partial}{\partial r} - \frac{1}{r} \sin \theta \frac{\partial}{\partial \theta} \right] F_1 + \left[\sin \theta \frac{\partial}{\partial r} + \frac{1}{r} \cos \theta \frac{\partial}{\partial \theta} \right] F_2$$

$$= \left[\cos \theta \frac{\partial}{\partial r} F_1 + \sin \theta \frac{\partial}{\partial r} F_2 \right] + \frac{1}{r} \left[-\sin \theta \frac{\partial}{\partial \theta} F_1 + \cos \theta \frac{\partial}{\partial \theta} F_2 \right]$$

$$= \underbrace{\left[\frac{\partial}{\partial r} F_1 \cos \theta + \frac{\partial}{\partial r} F_2 \sin \theta \right]}_{\frac{\partial}{\partial r} F_r} + \underbrace{\frac{1}{r} \left[-\sin \theta \frac{\partial}{\partial \theta} F_1 + \cos \theta \frac{\partial}{\partial \theta} F_2 \right]}_{\text{Not quite } \frac{1}{r} \frac{\partial}{\partial \theta} F_\theta}$$

$$\frac{\partial}{\partial r} F_r$$

$$\text{Not quite } \frac{1}{r} \frac{\partial}{\partial \theta} F_\theta = \frac{1}{r} \left[\frac{\partial}{\partial \theta} (-F_1 \sin \theta) + \frac{\partial}{\partial \theta} (F_2 \cos \theta) \right]$$

$$= \frac{1}{r} \left[F_1 (-\cos \theta) - F_2 (\sin \theta) - \sin \theta \frac{\partial}{\partial \theta} F_1 + \cos \theta \frac{\partial}{\partial \theta} F_2 \right]$$

$$\frac{\partial}{\partial r} F_r$$

Not quite $\frac{1}{r} \frac{\partial}{\partial \theta} F_\theta = \frac{1}{r} \left[\frac{\partial}{\partial \theta} (F_\theta \sin \theta) + \frac{\partial}{\partial \theta} (F_\theta \cos \theta) \right]$

$$= \frac{1}{r} \left[\underbrace{F_1(-\cos \theta) - F_2(\sin \theta)}_{\text{add } \frac{1}{2} \text{ subtract}} - \sin \theta \frac{\partial}{\partial \theta} F_1 + \cos \theta \frac{\partial}{\partial \theta} F_2 \right]$$

$$= \frac{\partial}{\partial r} F_r + \underbrace{\frac{1}{r} F_1 \cos \theta + \frac{1}{r} F_2 \sin \theta}_{\frac{1}{r} (F_r)} - \underbrace{\frac{1}{r} F_1 \cos \theta - \frac{1}{r} F_2 \sin \theta + \frac{1}{r} \left[-\sin \theta \frac{\partial}{\partial \theta} F_1 + \cos \theta \frac{\partial}{\partial \theta} F_2 \right]}_{= \frac{1}{r} \frac{\partial}{\partial \theta} F_\theta}$$

$$= \underbrace{\frac{\partial}{\partial r} F_r + \frac{1}{r} F_r}_{\text{Combine}} + \frac{1}{r} \frac{\partial}{\partial \theta} F_\theta$$

$$= \frac{1}{r} \frac{\partial}{\partial r} (r F_r) + \frac{1}{r} \frac{\partial}{\partial \theta} F_\theta$$

In terms of lamé coefficients $h_1=1$, $h_2=r$:

$$\vec{\nabla} \cdot \vec{F} = \frac{1}{(h_1 \cdot h_2)} \frac{\partial}{\partial r} (h_2 F_r) + \frac{1}{(h_1 \cdot h_2)} \frac{\partial}{\partial \theta} (h_1 F_\theta)$$

e.g. for $\vec{F} = \begin{bmatrix} x^2 + y^2 \\ -2y \end{bmatrix}$

In Cartesian Coordinates:

$$\vec{\nabla} \cdot \vec{F} = \frac{\partial}{\partial x} (x^2 + y^2) + \frac{\partial}{\partial y} (-2y) = 2x - 2$$

In polar Coordinates:

$$\vec{F} = \underbrace{(r^2 \cos \theta - 2r \sin^2 \theta)}_{F_r} \hat{e}_r - \underbrace{(r^2 \sin \theta + 2r \sin \theta \cos \theta)}_{F_\theta} \hat{e}_\theta$$

$$\text{Thus: } \vec{\nabla} \cdot \vec{F} = \frac{1}{r} \frac{\partial}{\partial r} (r F_r) + \frac{1}{r} \frac{\partial}{\partial \theta} (F_\theta)$$

$$= \frac{1}{r} \frac{\partial}{\partial r} (r [r^2 \cos \theta - 2r \sin^2 \theta]) + \frac{1}{r} \frac{\partial}{\partial \theta} (-r^2 \sin \theta - 2r \sin \theta \cos \theta)$$

$$= \frac{1}{r} [3r^2 \cos \theta - 4r \sin^2 \theta] + \frac{1}{r} (-r) [r \cos \theta + 2 \cos^2 \theta - 2 \sin^2 \theta]$$

$$= 3r \cos \theta - 4 \sin^2 \theta - r \cos \theta - 2 \cos^2 \theta + 2 \sin^2 \theta$$

$$= 2r \cos \theta - 2(\sin^2 \theta + \cos^2 \theta)$$

$$= \underbrace{2r \cos \theta - 2}$$

Same as $2x - 2$
